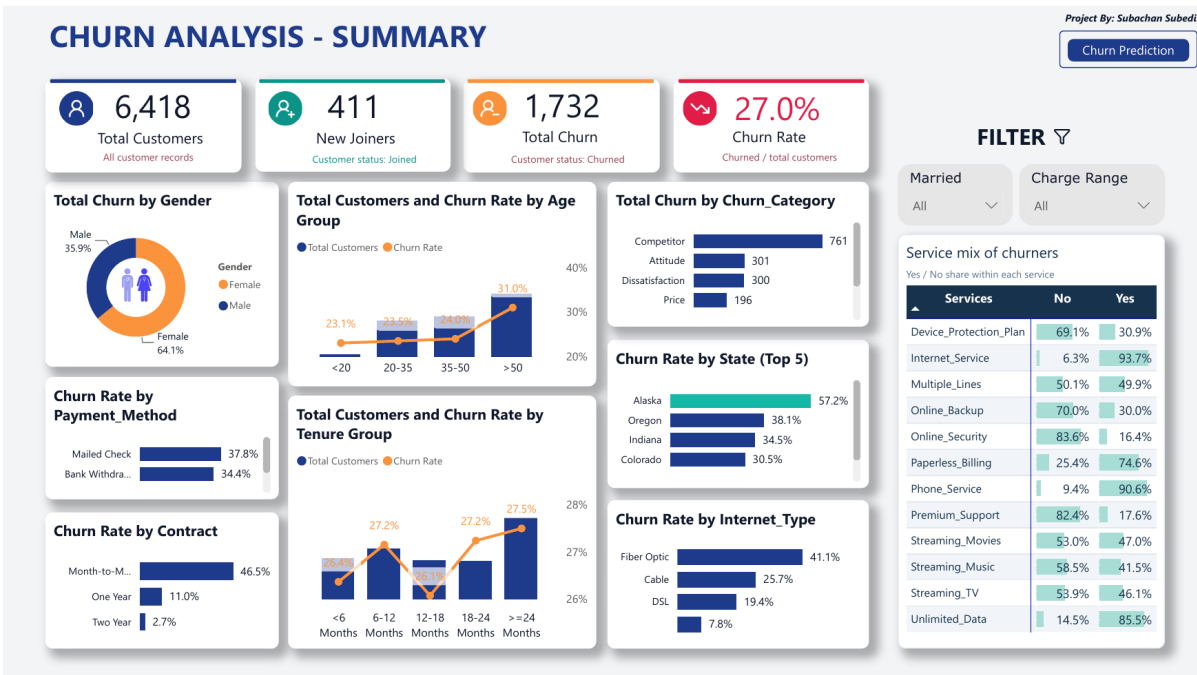


Customer Churn Analysis

Management report, dashboard guide, and implementation reference



Final Power BI summary dashboard

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Management, technical, and operating reference

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How to use this report

Sections 1 to 4 explain the business purpose and dashboard. Sections 5 to 9 document the dataset, SQL route, CSV route, model, and measures. Sections 10 to 13 interpret the results. Sections 14 to 16 provide controls, limitations, and handoff guidance.

1. Executive summary

Purpose and conclusion

This project converts customer, contract, service, financial, and churn records into a two-page Power BI report for retention analysis. The report shows the scale of churn, identifies where it is concentrated, explains the recorded reasons, and provides an imported customer list for structured review.

The clearest verified concern is the month-to-month contract group. Its churn rate is 46.5%, and it contains 1,529 of the 1,732 churned customers. Competitor is the largest recorded churn category with 761 customers. Fiber Optic customers also show a 41.1% churn rate, higher than Cable at 25.7% and DSL at 19.4%.

Headline metrics

Metric	Result	Management meaning
Customer records	6,418	Population represented in prod_Churn
Stayed	4,275	Customers retained in the supplied data
Churned	1,732	Customers marked Churned
Joined	411	New joiners included in the customer table
Overall churn rate	27.0%	Churned divided by all 6,418 records
Prediction records	381	Imported records available for review

Management priorities

1. Investigate the month-to-month population first, then segment it by internet type, tenure, and recorded churn category.
2. Review competitor-related reasons and fiber-optic service experience before designing a retention action.
3. Treat the Prediction page as a review list until the scoring method, forecast horizon, and evaluation results are documented.

2. Project overview and objectives

Business context

Customer churn reduces the active customer base and may weaken recurring revenue. Managers need more than a single churn percentage: they need to know which customer groups contribute most to churn, which reasons are recorded, how service choices are represented, and which records should be examined in detail.

Project objectives

- 1. Create a concise management view of customer volume, new joiners, churn volume, and churn rate.
- 2. Compare churn across customer profile, tenure, contract, payment method, internet type, state, and recorded reason.
- 3. Transform service columns into a structure that supports one consistent service matrix.
- 4. Provide a separate page for reviewing imported prediction records at customer level.
- 5. Document two reproducible preparation routes: SQL Server and CSV files with Power Query.

Intended readers

Reader	Primary need
Management	Understand concentration, prioritize investigation, and interpret KPIs correctly.
Retention or service teams	Identify segments and records for further review.
BI or data analysts	Understand transformations, measures, relationships, and refresh controls.
HR and portfolio reviewers	Evaluate the business reasoning, technical delivery, and documentation quality.

3. Business requirements and KPI definitions

Questions the dashboard must answer

Question	Dashboard response
How large is the customer population?	Total customers and customer-status counts.
How much churn is present?	Total churn and overall churn rate.
Where is churn concentrated?	Contract, age, tenure, payment, state, internet type, and gender views.
Why did customers leave?	Recorded churn category and churn reason fields.
Which services are common among churners?	Service matrix using unpivoted service records.
Which records require review?	Prediction filters, demographic breakdowns, and customer detail table.

KPI glossary

KPI	Definition	Model expression or rule
Total Customers	Count of Customer_ID in prod_Churn.	COUNT(prod_Churn[Customer_ID])
New Joiners	Customer records where status is Joined.	Filtered count of Customer_ID
Total Churn	Sum of the 1/0 Churn Status field.	SUM(prod_Churn[Churn Status])
Churn Rate	Churned records divided by all customer records.	[Total Churn] / [Total Customers]
Predicted Churner Count	Count of Customer_ID in Predictions.	COUNT(Predictions[Customer_ID]) + 0
Service mix	Yes and No shares within churned service records.	Composition measure, not service-level churn rate

Required report behavior

Slicers and chart selections should filter related visuals, navigation buttons should move between report pages, and the customer detail table should remain available for record-level review. Headline totals should reconcile after every refresh.

4. Dashboard overview

Report structure

Page	Purpose	Core components
Summary	Explain overall churn and segment concentration.	Four KPIs, nine analytical visuals, two slicers, and a service matrix.
Prediction	Review imported records and their distribution.	Prediction KPIs, six breakdown visuals, search and dropdown filters, and a detail table.

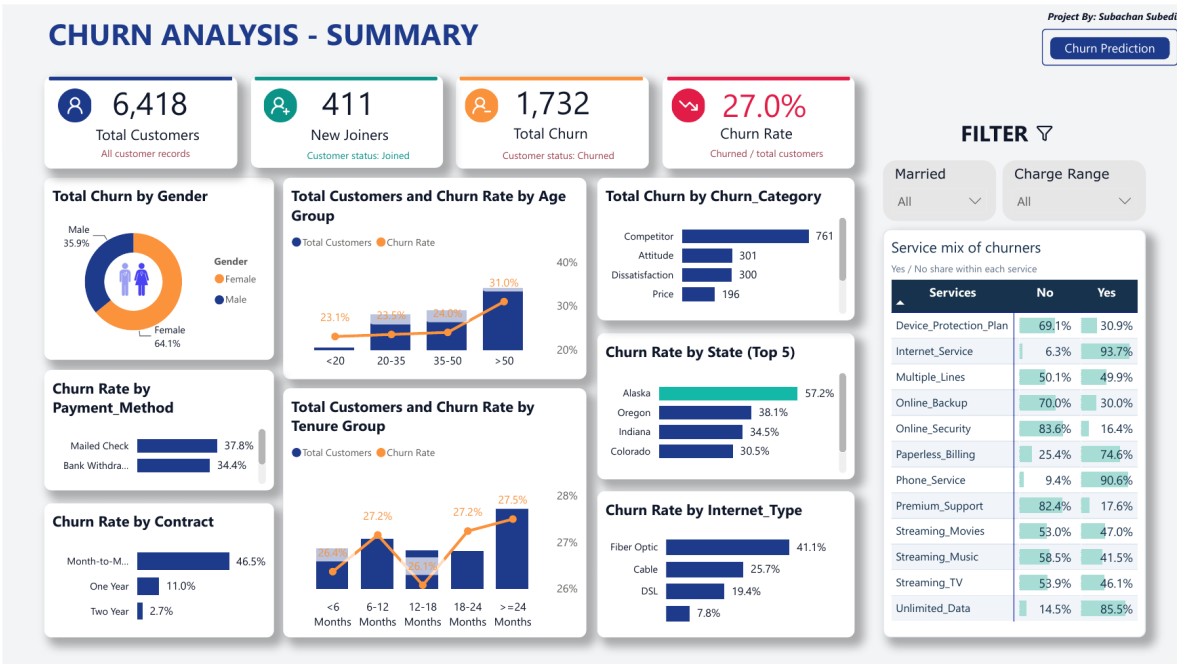


Figure 1. Summary page of the final Power BI report

Navigation and interaction

The Summary page links to Prediction, and Prediction links back to Summary. The report uses compact slicers, cross-filtering, and record-level tables so a reader can move from headline performance to a selected segment and then to the underlying records.

Reading order

1. Confirm the headline totals with all filters cleared.
2. Compare segment churn rates and volumes before focusing on a category.
3. Use the service matrix for churner composition, not for service-level churn risk.
4. Use the Prediction page only for the imported review population.

5. Dataset scope and table reference

Main customer table

prod_Churn contains 6,418 customer-level records and a unique Customer_ID. The table combines customer profile, relationship, service, financial, status, and recorded churn fields.

Field group	Verified fields in the project
Identifier	Customer_ID
Customer profile	Gender, Age, Married, State, Number_of_Referrals
Relationship	Tenure_in_Months, Contract, Payment_Method, Paperless_Billing
Services	Phone_Service, Multiple_Lines, Internet_Service, Internet_Type, Online_Security, Online_Backup, Device_Protection_Plan, Premium_Support, Streaming_TV, Streaming_Movies, Streaming_Music, Unlimited_Data
Financial	Monthly_Charge, Total_Charges, Total_Refunds, Total_Extra_Data_Charges, Total_Long_Distance_Charges, Total_Revenue
Outcome	Customer_Status, Churn_Category, Churn_Reason, derived Churn Status

Derived and supporting tables

Table	Grain and role	Verified size
prod_Services	One customer-service response per row after unpivoting 12 service columns.	77,016 rows
Predictions	One imported customer record per Customer_ID for review.	381 rows
mapping_AgeGrp	Age group labels and numeric sort order.	Supporting table
mapping_TenureGrp	Tenure group labels and numeric sort order.	Supporting table
tbl_Measures	Dedicated location for report measures.	Six measures

The main table reconciles to 4,275 Stayed, 1,732 Churned, and 411 Joined records. These three statuses total 6,418 customer records.

6. Solution architecture and model

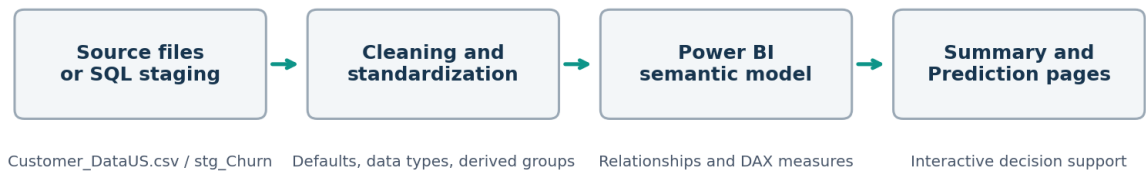


Figure 2. Verified project delivery flow

Model relationships

Relationship	State	Purpose
prod_Churn to mapping_AgeGrp	Active, many to one	Apply ordered age groups to the main customer table.
prod_Churn to mapping_TenureGrp	Active, many to one	Apply ordered tenure groups.
prod_Services to prod_Churn	Active	Filter service responses through the customer record.
Predictions to prod_Churn	Active, one to one, bidirectional	Connect imported review records with customer attributes.
Predictions to age and tenure mappings	Inactive	Avoid competing filter paths in the active model.

Design rationale

The customer table remains the central analytical entity. Age and tenure mappings preserve business-friendly category order. The service table separates repeated service attributes from the wide customer row, allowing one matrix to compare all service fields. Measures are isolated in a dedicated table to simplify maintenance.

Source flexibility

The same reporting model can be fed through SQL Server objects or through CSV files prepared in Power Query. The final PBIX uses the CSV route, while the SQL route remains a documented alternative for centralized data management.

7. SQL Server preparation workflow

Workflow

Stage	Action	Output or control
1. Database	Create the project database.	db_Churn
2. Staging import	Load the source CSV using the Flat File Import Wizard.	stg_Churn
3. Key and null settings	Use Customer_ID as primary key; allow nulls in other imported columns.	Stable staging load
4. Exploration	Review Gender, Contract, Customer_Status, and State distributions.	Distribution checks
5. Null audit	Count nulls across customer, service, financial, and churn fields.	Data-quality baseline
6. Production load	Select cleaned fields and replace selected nulls.	prod_Churn
7. Reporting views	Separate Churned/Stayed from Joined records.	vw_ChurnData; vw_JoinData
8. Power BI load	Connect Power BI to SQL Server objects.	Refreshable import model

Core SQL objects

```
CREATE DATABASE db_Churn;
SELECT ... INTO db_Churn.dbo.prod_Churn FROM db_Churn.dbo.stg_Churn;
CREATE VIEW vw_ChurnData AS SELECT * FROM prod_Churn WHERE Customer_Status IN ('Churned','Stayed');
CREATE VIEW vw_JoinData AS SELECT * FROM prod_Churn WHERE Customer_Status = 'Joined';
```

Cleaning rules supplied with the project

Selected missing service fields are replaced with No or None. Missing churn category and churn reason values are replaced with Others. The import guidance uses text types for Yes and No fields when the wizard incorrectly infers Bit.

8. CSV and Power Query workflow

Step	Power Query operation	Result
1	Read CSV with comma delimiter, UTF-8 encoding, and CSV quote handling.	Source rows loaded
2	Promote the first row to headers.	Expected column names
3	Assign numeric types to age, tenure, charges, refunds, and revenue.	Reliable calculations
4	Standardize missing categorical values.	No, None, or Others as applicable
5	Add Churn Status and charge-range bands.	Reusable measures and slicers
6	Create age and tenure mapping queries with sort columns.	Ordered category labels
7	Unpivot 12 service columns.	prod_Services with 77,016 rows
8	Load PredictionsUS.csv using the expected header set.	381 imported records

Source and blank-handling patterns

```
Source = Csv.Document(File.Contents("...\Customer_DataUS.csv"), [Delimiter=",", Encoding=65001,
QuoteStyle=QuoteStyle.Csv])
Headers = Table.PromoteHeaders(Source, [PromoteAllScalars=true])
each if _ = null or Text.Trim(_) = "" then "None" else Text.Trim(_)
```

Why empty text can remain visible

Power Query treats null, an empty string, and whitespace as different values. A Replace Values step that targets only null does not change empty text. Cleaning must occur in the main customer query before the internet-type visual or downstream queries are built.

9. Power BI calculations and grouping logic

Verified measures

Measure	Definition	Use
Total Customers	COUNT(prod_Churn[Customer_ID])	Headline population and denominators
New Joiners	Filtered count where Customer_Status = Joined	Headline acquisition volume
Total Churn	SUM(prod_Churn[Churn Status])	Churn volume
Churn Rate	[Total Churn] / [Total Customers]	Overall and segmented rate
Count Predicted Churner	COUNT(Predictions[Customer_ID]) + 0	Prediction-page record count
Title Predicted Churner	Text joined to the prediction count	Dynamic detail-table title

Grouping boundaries

Group	Implemented boundaries	Display consideration
Age	Below 20; 20-34; 35-49; 50 or above	The displayed labels are less precise and should be aligned to the implemented boundaries.
Tenure	Below 6; 6-11; 12-17; 18-23; 24 or more months	Numeric sort columns preserve chronological order.
Monthly charge	Derived range field used by the Summary slicer	Refresh after numeric type assignment.

Denominator discipline

The model Churn Rate divides churned records by Total Customers. Because Total Customers includes Stayed, Churned, and Joined records, the 27.0% headline rate uses all 6,418 records as the denominator. This definition should remain explicit whenever the result is compared with another churn report.

Formatting and interaction

Measures are formatted for whole-number counts or percentages as appropriate. Cross-filtering and slicers change the filter context, so readers should clear selections before reconciling headline totals.

10. Summary dashboard analysis

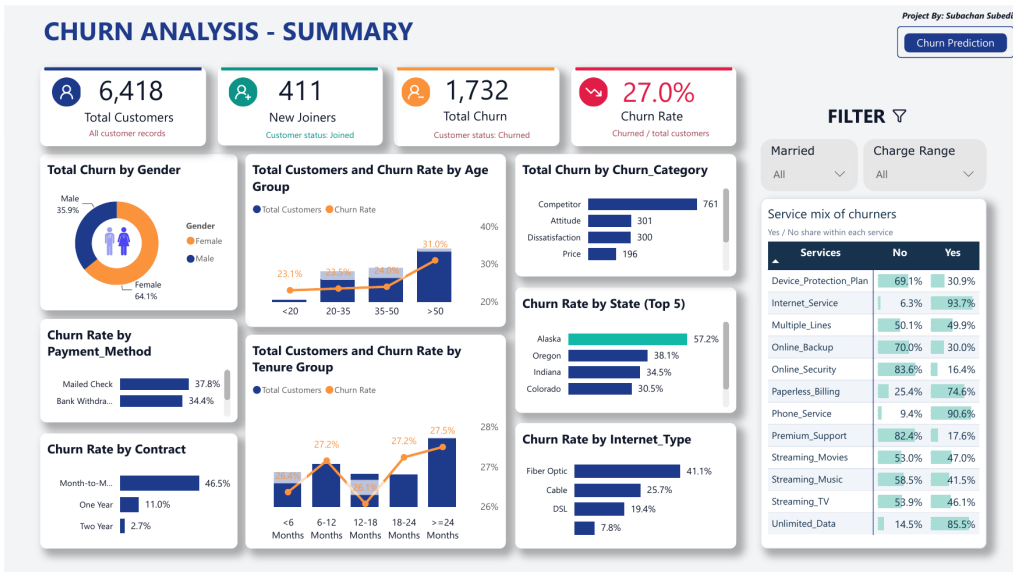


Figure 3. Final Summary page

How the page supports decisions

The KPI row establishes the population, joiners, churn volume, and headline rate. Supporting visuals compare customer profile, tenure, payment, contract, reasons, state, and internet type. The service matrix describes churner composition, while the slicers focus the page by marital status and monthly-charge range.

Contract pattern



Figure 4. Churn rate differs sharply by contract type

Month-to-month customers have both the highest churn rate and the largest churn volume. The contract result is the strongest starting point because it is supported by both rate and count, not by rate alone.

11. Prediction dashboard analysis

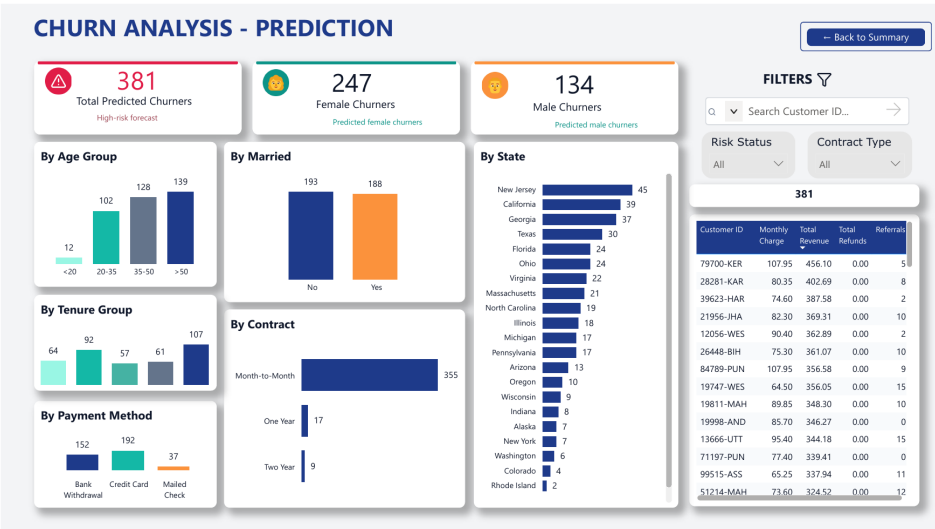


Figure 5. Final Prediction page

Imported review population

Breakdown	Records	Share of list
Month-to-Month	355	93.2%
One Year	17	4.5%
Two Year	9	2.4%

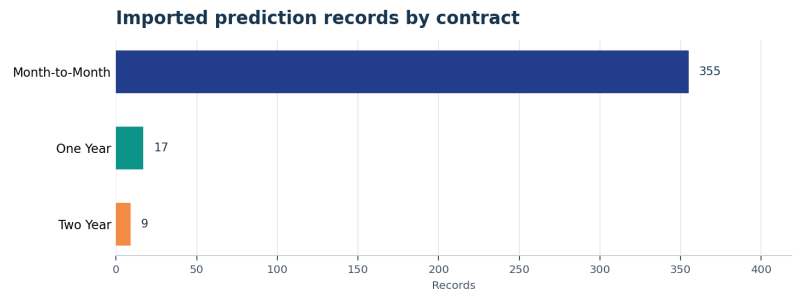


Figure 6. The imported review list is concentrated in month-to-month contracts

Interpretation

The list contains 247 female and 134 male records. All 381 records have Customer_Status_Predicted equal to 1. The project materials do not include training code, probabilities, forecast horizon, threshold selection, or evaluation results, so the page supports record review but does not establish predictive accuracy.

12. Key findings and interpretation

Recorded reasons

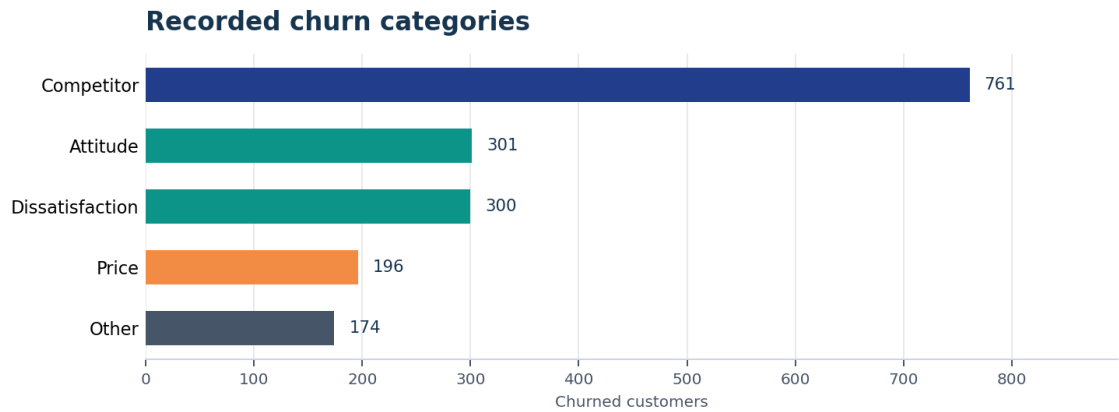


Figure 7. Competitor is the largest recorded churn category

Category	Churned customers	Share of 1,732
Competitor	761	43.9%
Attitude	301	17.4%
Dissatisfaction	300	17.3%
Price	196	11.3%
Other	174	10.0%

Service and demographic signals

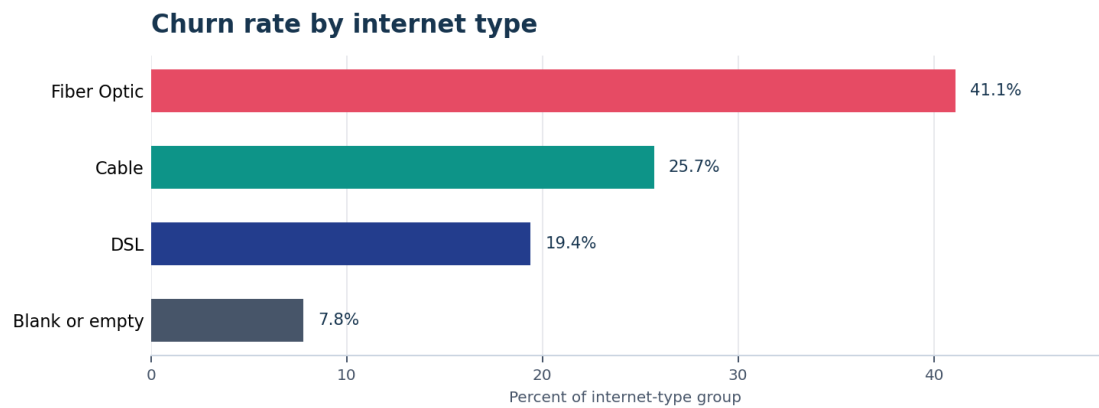


Figure 8. Fiber Optic has the highest displayed internet-type churn rate

The implemented 50-and-above age group has a 31.0% churn rate. These results identify segments for investigation, but they do not prove that age or internet type causes churn. Contract, tenure, service experience, and customer mix may overlap.

13. Management recommendations

Recommended action sequence

Priority	Action	Evidence to review before action
1. Contract concentration	Profile month-to-month churn by tenure, internet type, charge range, and churn category.	Volume, rate, and cross-filtered customer records
2. Competitor reasons	Review the detailed reasons behind the 761 Competitor-category records.	Churn_Reason and customer-level context
3. Fiber Optic experience	Compare service fields and recorded reasons within Fiber Optic customers.	Service mix, contract, tenure, and churn reason
4. Prediction governance	Document model source, scoring date, horizon, threshold, probability, and evaluation.	Scoring documentation and validation results
5. Outcome measurement	Define a baseline, action group, comparison group, and outcome window.	Pre-action churn and post-action retention results

Decision principles

- Prioritize categories that combine high rate and high volume.
- Use dashboard patterns to form questions, then inspect the underlying records before acting.
- Separate descriptive analysis from prediction evidence and from measured intervention results.
- Retain the existing report as a monitoring view, while documenting any new campaign or operational measures separately.

Expected business value

The project provides a repeatable way to identify retention priorities, communicate evidence to non-technical stakeholders, and move from headline churn to the segments and records behind it. Business impact should be reported only after a retention action and outcome-measurement process are implemented.

14. Data quality and measure observations

Definition issues that affect interpretation

Observation	Why it matters	Control
Joined records are in the churn-rate denominator.	The 27.0% rate differs from a rate calculated only on Stayed and Churned records.	State the denominator in the KPI definition.
Empty Internet_Type text can appear as a blank category.	Replacing null alone does not clean empty strings or whitespace.	Trim text and test null plus empty values in the main query.
Age labels do not exactly match implemented boundaries.	A label such as greater than 50 excludes age 50 in ordinary reading, while the logic includes it.	Rename labels to 50 or above and align all bins.
Service matrix values are churner composition.	Readers may mistake Yes/No shares for service-level churn rates.	Keep the subtitle and interpretation note.
Prediction evidence is incomplete.	A review list is not a validated predictive model.	Add scoring and evaluation documentation before operational use.
Prediction tenure labels are not visible in the supplied export.	The chart is harder to interpret outside the PBIX.	Enable category labels and confirm them in export.

Reconciliation checkpoints

Checkpoint	Expected result
Customer records	6,418
Stayed / Churned / Joined	4,275 / 1,732 / 411
Service rows	77,016
Prediction records	381
Month-to-month churned customers	1,529
Competitor-category churners	761

15. Refresh and reproduction guide

CSV route

1. Place Customer_Data.csv and Predictions.csv in an accessible location.
2. Open Transform data, select Data source settings, and update both file paths.
3. Confirm the expected headers, especially Customer_ID, before applying changes.
4. Apply the null, empty-text, and whitespace cleaning rule in the main customer query.
5. Refresh the model and reconcile the checkpoint totals below.

SQL Server route

1. Create db_Churn and import the customer CSV into stg_Churn.
2. Apply the primary key, null settings, data-type checks, distribution checks, and null audit.
3. Create prod_Churn and the two reporting views using the supplied logic.
4. Connect Power BI to SQL Server, select the intended objects, and refresh.

Post-refresh checklist

Control	Required verification
Schema	Expected columns are present and named consistently.
Data types	Age, tenure, charges, refunds, and revenue are numeric.
Categories	No unintended blank service or internet categories remain.
Counts	Customer, churn, service, and prediction totals reconcile.
Relationships	Active links remain valid and no ambiguity warning appears.
Visual behavior	Slicers, cross-filtering, navigation, search, and tables work.
Export review	Labels, scroll positions, and page content are readable in PDF.

16. Limitations and conclusion

Limitations

- The supplied materials do not establish a reporting period or currency.
- Revenue fields are present, but profit, campaign return, and revenue saved are not measured.
- The analysis is descriptive and does not establish causal relationships.
- The prediction method, forecast horizon, probabilities, threshold, and evaluation results are not supplied.

Conclusion

The final Power BI project brings customer churn, contract behavior, recorded reasons, service composition, and an imported review list into one coherent reporting experience. The strongest supported management priority is the month-to-month contract population, followed by competitor-related churn reasons and Fiber Optic service experience.

The project also demonstrates a complete BI delivery pattern: source ingestion, data-quality checks, transformation, dimensional support tables, DAX measures, interactive reporting, refresh controls, and documented interpretation limits. SQL Server provides a controlled centralized route, while CSV and Power Query provide a portable alternative.

Handoff statement

The dashboard is ready for analytical review using the supplied data. Before it is used for automated customer outreach or claims of retention impact, the prediction process and outcome-measurement framework should be documented and validated.